



TEMPORAL TIME PROCESSES

by Josh Job Joseph

INTRODUCTION

What is It?

Temporal Point Processes (TPP)
are probabilistic generative
models designed for continuous-
time event sequences.

Neural TPPs add deep learning in
order to make the models more
flexible and efficient.

What is it Used For?

A variety of applications such as modeling
earthquakes, predicting social media
behavior, earthquakes, stock behaviors, and
many other event-based situations in
which the occurrence of one event causes
the occurrence of other events.

SOURCE: "NEURAL TEMPORAL POINT PROCESSES: A REVIEW"

BY OLEKSANDR SHCHUR, ALI CANER TURKMEN, TIM JANUSCHOWSKI, STEPHAN GÜNNEMANN

COMMONLY USED POINT PROCESSES

Poisson Point Process

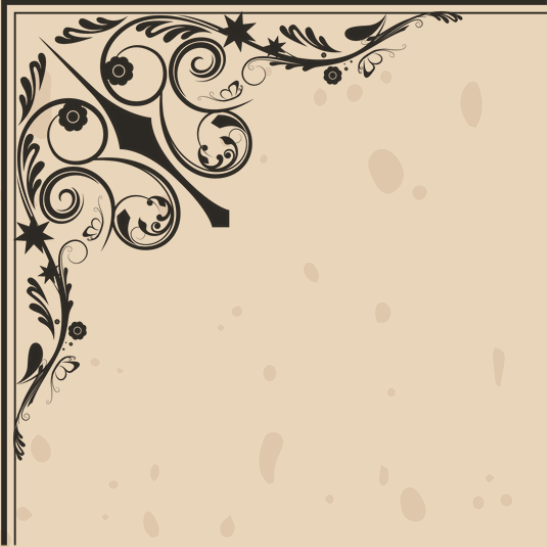
- Complete independence: the number of points of a Poisson Point Process in each bounded region will be completely independent of others.

Hawkes Point Processes

- Considered “de facto” tool to model event streams.
- The intensity function (typically static) is history-dependent, thus allowing it to model history-dependent events.

SOURCE: “TRANSFORMER HAWKES PROCESS”

BY SIMIAO ZUO, HAOMING JIANG, ZICHONG LI, TAO ZHAO AND HONGYUAN ZHA



HAWKES PROCESS



Common issue with the Hawkes Process:

- the intensity function is static, thus limiting the ability to capture complicated dynamics.
- Assumes that all past events should have positive influences on the occurrence of current events.

RNN with Hawkes Process:

- Allows the model to be able summarize historic events either in discrete or continuous time using recurrent structures.
- This allows for the model to not treat all historical contributions as strictly additive.
- More complex memory effects such as delay can be used

SOURCE: "SELF-ATTENTIVE HAWKES PROCESS"

BY QIANG ZHANG, ALDO SIPANI, OMER KIRNAP, EMINE YILMAZ



HAWKES PROCESS KERNELS

Exponential Kernel

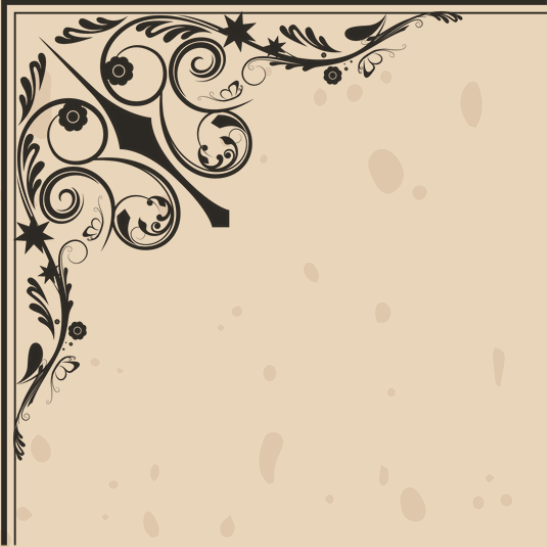
- Models how the event decays overtime
- Makes future events less likely to occur as time goes on since events are tightly clustered together.

Power Law Kernels

- Kernels are used when the influence of an event decays more slowly overtime.
- Better for events that have a long lasting impact/effect

SOURCE: "A BRIEF INTRODUCTION TO HAWKES PROCESSES"

BY HEY AMIT



CONDITIONAL INTENSITY



What is it?

- Time dependent rate, $\lambda(t)$, that tells you the probability of an event occurring at a given moment in time, based on past events.
- “Backbone” of the Hawkes Process.
- Constantly updating based on the occurrence of new events.



SOURCE: “A BRIEF INTRODUCTION TO HAWKES PROCESSES”

BY HEY AMIT



LIKELIHOOD ESTIMATION

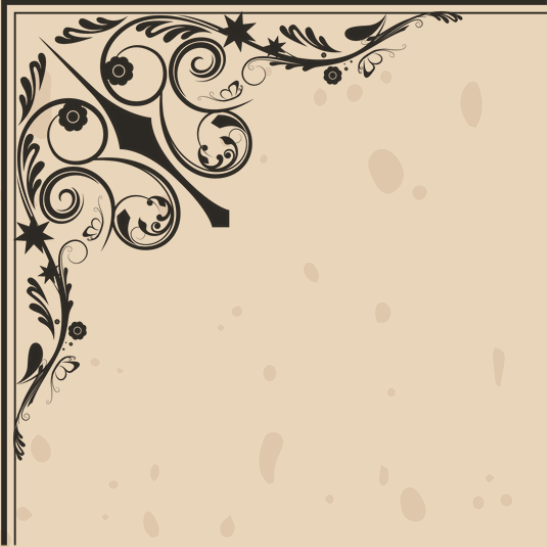
Maximum Likelihood Estimation

- Estimates the parameters of the Hawkes Process such as: baseline intensity μ and the triggering function $\phi(t)$
- Based on data so that the model fits the data

Bayesian Inference

- Estimates parameters based on observed data and prior beliefs
- Better for situations where you are working with noisy data or limited data.

SOURCE: "A BRIEF INTRODUCTION TO HAWKES PROCESSES"
BY HEY AMIT



PYTHON LIBRARIES



Tick

Considered to be the most popular library for implementing Hawkes Processes. Not only this, but it also includes multivariate tools. It supports both exponential and power-law kernels, which gives additional flexibility for tuning the model for different types of self-exciting behavior.

- Fast due use of sparse arrays
- Has MLE natively supported



SOURCE: “A BRIEF INTRODUCTION TO HAWKES PROCESSES”

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THANK YOU